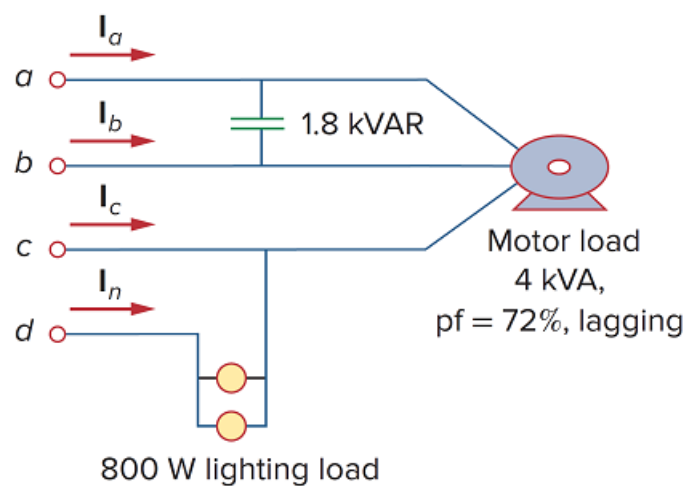


Problem

Figure 12.76 displays a three-phase delta-connected motor load which is connected to a line voltage of 440 V and draws 4 kVA at a power factor of 72 per cent lagging. In addition, a single 1.8 kVAR capacitor is connected between lines *a* and *b*, while a 800-W lighting load is connected between line *c* and neutral. Assuming the *abc* sequence and taking

$V_{an} = V_p \angle 0^\circ$, find the magnitude and phase angle of currents I_a , I_b , I_c , and I_n .

Figure 12.76



Step-by-step solution

Step 1 of 4

We first find the magnitude of various currents.

For motor,

$$I_L = \frac{S}{\sqrt{3}V_L} = \frac{4000}{440\sqrt{3}} = 5.248 \text{ A}$$

For capacitor,

$$I_C = \frac{Q_C}{V_C} = \frac{1800}{440} = 4.091 \text{ A}$$

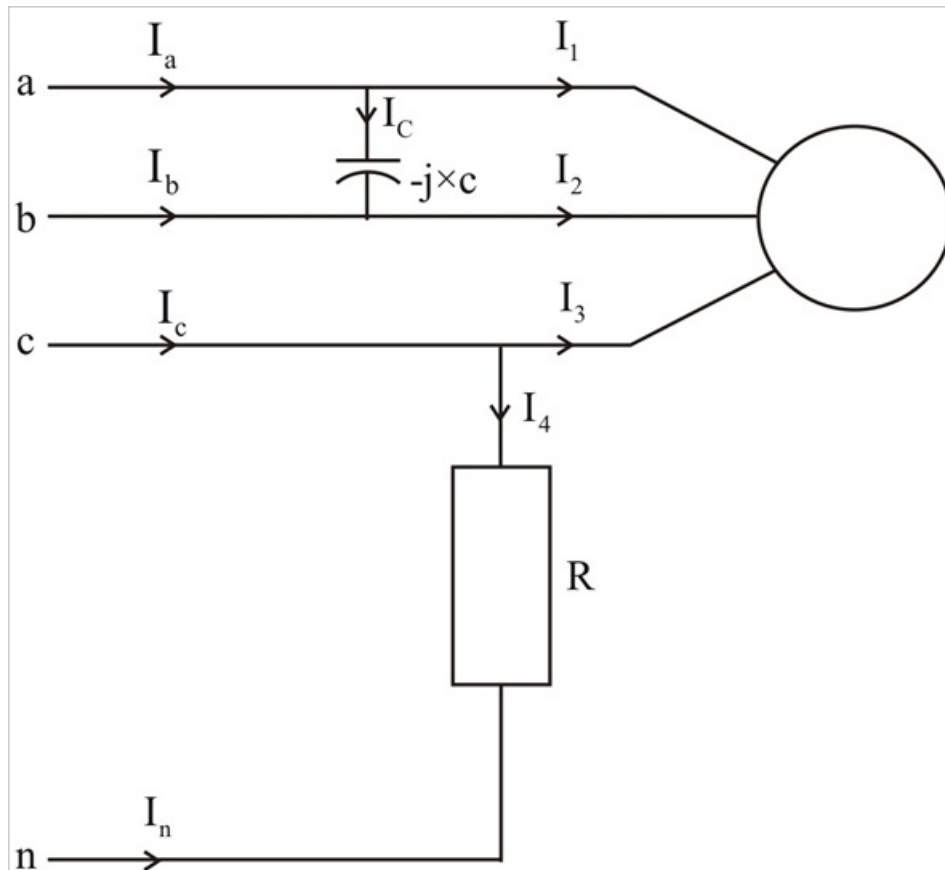
For the lighting,

$$V_p = \frac{440}{j3} = 254 \text{ V}$$

$$I_{L_1} = \frac{P_{L_1}}{V_p} = \frac{800}{254} = 3.15 \text{ A}$$

Step 2 of 4

Considering the figure below.



Step 3 of 4

If

$$V_{ah} = V_p \angle 0^\circ$$

$$V_{ab} = \sqrt{3} V_p \angle 30^\circ$$

$$V_{ch} = V_p \angle 120^\circ$$

$$I_C = \frac{V_{ab}}{-j \times C} = 4.091 \angle 120^\circ$$

Where,

$$\theta = \cos^{-1}(0.72) = 43.95^\circ$$

$$I_1 = 5.249 \angle 73.95^\circ$$

$$I_2 = 5.249 \angle -46.03^\circ$$

$$I_3 = 5.249 \angle 193.95^\circ$$

$$I_{L_1} = \frac{V_{C_N}}{R} = 3.18 \angle 120^\circ$$

Step 4 of 4

Thus,

$$\begin{aligned} I_a &= I_1 + I_C = 5.249 \angle 73.95^\circ + 4.091 \angle 120^\circ \\ &= 8.608 \angle 93.96^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} I_b &= I_2 - I_C = 5.249 \angle -46.05^\circ - 4.091 \angle 120^\circ \\ &= 9.271 \angle -52.16^\circ \text{ A} \end{aligned}$$

$$\begin{aligned} I_C &= I_3 + I_{L_3} = 5.249 \angle 193.95^\circ + 3.15 \angle 120^\circ \\ &= 6.827 \angle 167.6^\circ \text{ A} \end{aligned}$$

$$I_n = -I_{L_3} = 3.15 \angle -60^\circ \text{ A} .$$